SYSC4005/5001

Simulation & Modelling Project 1 Report

Two-Node Computer Processing System

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Simulation Parameters

Parameter	Value
Replications	30
Simulation time per run	5,000,000 ms (5,000 s)
Confidence level	95% ($t_{\square-1}$ = 2.045)
Buffer 1 mean IAT	4 ms (Exponential)
Buffer 2 mean IAT	12 ms (Exponential)
N1 service – Type I	Uniform [1, 3] ms
N1 service – Type II	Uniform [2, 6] ms
N2 service (each proc.)	Exponential, mean 5 ms
N1 slot B1 / B2	50 ms / 30 ms
Router 1 redirect threshold	> 5 packets in N2 queue
Type-II forward probability	0.5

1. System Description

The simulated system is a two-node computer processing network handling two classes of packet traffic. Packets arrive at two separate buffers at Processing Node 1, are processed by a single alternating server, then routed (or discarded) through Router 1 before possibly joining the queue at Processing Node 2.

1.1 Arrival Process

Both buffer arrival streams are Poisson processes, meaning inter-arrival times are independently and identically distributed (i.i.d.) exponential random variables.

- Buffer 1 (Type **I** packets): mean inter-arrival time = 4 ms ⇒ arrival rate $\lambda_1 = 0.25$ pkt/ms
- Buffer 2 (Type **II** packets): mean inter-arrival time = 12 ms ⇒ arrival rate $\lambda_2 \approx 0.0833$ pkt/ms

1.2 Processing Node 1

Node 1 employs a single processor that cycles between the two buffers on a time-slotted basis:

- Serves Buffer 1 for a 50 ms time slot, then switches to Buffer 2 for 30 ms, repeating indefinitely.
- If the scheduled buffer is empty, the processor continues serving the other buffer without waiting.
- Once a packet's service begins, it runs to completion before any slot switch.
- Service times are uniformly distributed: Type **I** ~ Uniform[1, 3] ms (mean 2 ms), Type **II** ~ Uniform[2, 6] ms (mean 4 ms).

1.3 Router 1

After a packet departs Node 1, Router 1 decides its fate. The router operates instantaneously (zero delay):

- Type **I** packets are always forwarded toward Node 2.
- Type **II** packets are forwarded to Node 2 with probability 0.5, or discarded with probability 0.5.
- A packet forwarded by the routing logic is then inspected by Router 1: if more than 5 packets are waiting in the Node 2 buffer, the packet is redirected (dropped from the main path).

1.4 Processing Node 2

Node 2 has two parallel identical processors sharing a common queue (M/M/2-style station). Processing times are exponentially distributed with a mean of 5 ms each. An arriving packet immediately occupies a free processor if one is available, otherwise, it waits in the shared queue.

2. Simulation Methodology

2.1 Event-Driven Architecture

The simulation uses a discrete-event paradigm. A global priority queue (min-heap) stores future events ordered by scheduled time. The simulation clock advances to each successive event, and state changes are processed immediately. Four event types are used:

- EVT_ARRIVE_B1: A new Type-I packet joins Buffer 1, and the next arrival is immediately scheduled.
- EVT_ARRIVE_B2: A new Type-II packet joins Buffer 2, and the next arrival is immediately scheduled.
- EVT_N1_DONE: Node 1 finishes serving a packet, routing and next-service selection follow.
- EVT_N2_DONE: A Node 2 processor completes service, and the next queued packet (if any) is dispatched.

2.2 Time-Average Queue-Length Estimation (PASTA / Little's Law)

The average number of packets in each station is computed via the area-under-the-curve (AUC) method. Before processing each event at time t_k, the elapsed interval Δt = t_k − t_{k-1} is multiplied by the current queue length L(t_{k-1}) and accumulated:

```
area += Δt × L(t_prev)
```

At the end of the simulation run, the time-average is obtained by dividing the accumulated area by the total simulation time T:

```
E[L] = area / T
```

This is a direct application of the time-average formula for stationary processes. By Little’s Law, E[L] = λ × E[W], linking average queue length to average wait time and throughput.

For questions (a) and (b), the average system population (queue + in-service) for each type is tracked separately. Two additional area accumulators (areaN1InSvcT1 and areaN1InSvcT2) record time spent serving a Type-I and Type-II packet, respectively:

```
if (n1InSvc == 1) {
    if (n1CurrentType == 1)  st.areaN1InSvcT1 += dt;
    else                     st.areaN1InSvcT2 += dt;
}
st.avgSystemB1 = (st.areaB1 + st.areaN1InSvcT1) / SIM_TIME;
st.avgSystemB2 = (st.areaB2 + st.areaN1InSvcT2) / SIM_TIME;
```

2.3 Slot-Switching Logic (tryStartN1v2)

The most subtle part of the simulation is correctly managing Node 1's alternating slot schedule. The key insight is that idle time must consume the current slot's budget, otherwise the processor would unfairly extend a slot through idle periods.

A wall-clock slot expiry deadline (n1SlotExpiry) is maintained as an absolute timestamp. When Node 1 becomes free, a while-loop advances the slot expiry from its previous value until it lies in the future:

```
while (clock >= slotExpiry) {
    activeBuffer = (activeBuffer == 1) ? 2 : 1;
    slotExpiry  += (activeBuffer == 1 ? SLOT_B1 : SLOT_B2);
}
```

This ensures that if Node 1 was idle for, say, 65 ms, all elapsed 50-ms and 30-ms slots are properly expired before the next packet is scheduled, and the processor begins serving from the correct buffer.

2.4 Confidence Interval Calculation

Thirty independent replications are executed, each seeded with a distinct RNG seed. For each metric m, the per-replication means m_1, …, m_30 form a sample. The 95% confidence interval is constructed using the Student's t-distribution with n−1 = 29 degrees of freedom (critical value t* = 2.045):

$$CI = \bar{x} \pm t^* \times (s / \sqrt{n}) \qquad \text{where } s = \text{sample standard deviation, } n = 30$$

Each replication runs for 5,000,000 ms of simulated time, long enough to give stable estimates and keep half-widths below 0.5% of the point estimate for all metrics.

3. Simulation Results

All results below are 95% confidence intervals computed across 30 independent replications of 5,000,000 ms each.

Metric	Mean	± Half-width	95% CI
(a) Avg # Type-I pkts in Buffer1 + being served by N1	2.03611	0.00342	[2.03270, 2.03953]
(b) Avg # Type-II pkts in Buffer2 + being served by N1	1.14748	0.00262	[1.14486, 1.15010]
(c) Avg Number Type-I packets that waited in Buffer 1	1.53594	0.00341	[1.53253, 1.53936]
(d) Avg Number Type-I packets that waited in Buffer 2	0.81430	0.00031	[0.81179, 0.81682]
(e) Avg wait time – Buffer 1 (ms)	6.142	0.01386	[6.12814, 6.15586]
(e) Avg wait time – Buffer 2 (ms)	9.777	0.02774	[9.74903, 9.80451]
(f) Type-I sojourn at N1 (ms)	8.142	0.01388	[8.12824, 8.15600]
(f) Type-II sojourn at N1 (ms)	13.777	0.02777	[13.74924, 13.80478]
(g) Sojourn at N2 station (ms)	8.531	0.00715	[8.52339, 8.53770]
(h) P(packet redirected by Router 1)	0.01982	0.00009	[0.01973, 0.01991]

3.1 Average System Population at Node 1 – Questions (a) and (b)

(a) Average number of Type-I packets in Buffer 1 and in service at N1: 2.036 ± 0.003 packets.

This is calculated as the time-average of (queue length of B1) + (indicator that N1 is currently serving a Type-I packet). Using the AUC method:

```
avgSystemB1 = (areaB1 + areaN1InSvcT1) / SIM_TIME
```

With $\lambda_1 = 0.25$ pkt/ms and mean service time at N1 for Type-I of 2 ms, the utilization contributed by Type-I traffic is roughly 0.25×2 = 0.5 (but the alternating schedule modifies this). Little's Law confirms: $E[L] \approx \lambda \times E[W] = 0.25 \times 8.14 \approx 2.04$, consistent with the simulation result.

(b) Average number of Type-II packets in Buffer 2 and in service at N1: 1.147 ± 0.003 packets.

Computed symmetrically using areaN1InSvcT2. Little's Law check: $\lambda_2 \times E[W_2] = (1/12) \times 13.78 \approx 1.15$, matching the simulation.

3.2 Average Number of Packets That Had to Wait – Questions (c) and (d)

For each packet, the wait in buffer = serviceStartN1 − arrivalTime. A packet 'waited' if this value exceeds a numerical threshold of 1e-9 ms (to avoid floating-point artifacts from packets served immediately on arrival).

(c) Avg Number Type-I packets that waited in Buffer 1: 1.53594 ± 0.00341, 95% CI: [1.53253, 1.53936]

(d) Avg Number Type-II packets that waited in Buffer 2: 0.81430 ± 0.00251, 95% CI: [0.81179, 0.81682]

These values represent the expected number of packets waiting in each buffer at an arbitrary point in time.

Consistency check using Little's Law

Using Little's Law on the waiting portion:

$E[L_q] = \lambda \cdot E[W_q]$

- Buffer 1: 0.25 × 6.14 ≈ 1.540.25
- Buffer 2: (1/12) × 9.78 ≈ 0.82

This confirms consistency with the simulation.

For completeness, we also report the fraction of packets that experienced non-zero waiting:

- Type-I: ~83.34%
- Type-II: ~83.33%

This represents the probability that an arriving packet must wait before service, which complements the time-average queue length.

3.3 Average Waiting Times in Buffers – Question (e)

Waiting time is the duration a packet spends in its buffer before service begins at Node 1. It is recorded per-packet and averaged across all served packets:

```
waitInBuffer = packet.serviceStartN1 - packet.arrivalTime
```

Buffer 1 (Type I): 6.142 ± 0.014 ms | 95% CI: [6.128, 6.156] ms

Buffer 2 (Type II): 9.777 ± 0.028 ms | 95% CI: [9.749, 9.805] ms

Buffer 2 has a longer average wait despite having a lower arrival rate, because Type-II packets receive less server attention (30 ms slots vs 50 ms slots) and their service times are longer (mean 4 ms vs 2 ms for Type-I).

3.4 Sojourn Time at Node 1 – Question (f)

Sojourn time at N1 = wait in buffer + service time at N1. It is computed per packet as:

```
sojourn1 = packet.departN1 - packet.arrivalTime
```

Type-I sojourn at N1: 8.142 ± 0.014 ms | 95% CI: [8.128, 8.156] ms

Type-II sojourn at N1: 13.777 ± 0.028 ms | 95% CI: [13.749, 13.805] ms

Type-I sojourn = avg wait (6.14 ms) + mean service (2 ms) = 8.14 ms. Type-II = 9.78 + 4.0 = 13.78 ms. These figures agree precisely with the simulation, providing strong evidence that the accounting is correct.

3.5 Sojourn Time at Node 2 Station – Question (g)

Sojourn at N2 covers the entire time a packet spends in the Node 2 station, both waiting in the common buffer and being served by one of the two processors:

```
sojournN2 = packet.departN2 - packet.arrivalN2
```

N2 station sojourn: 8.531 ± 0.007 ms | 95% CI: [8.523, 8.538] ms

With two parallel Exp(5 ms) servers, the theoretical M/M/2 mean service rate per server is $\mu = 0.2$ pkt/ms. The effective arrival rate into N2 is the throughput from N1 after routing (most Type-I packets plus ~50% of Type-II packets, minus redirects). The narrow CI (±0.007 ms) reflects high precision from long runs.

3.6 Redirect Probability at Router 1 – Question (h)

Router 1 redirects a packet when the N2 queue length strictly exceeds 5 at the moment of arrival. The redirect probability is computed as:

```
pRedirect = redirectedCount / (servedByN2 + redirectedCount)
```

P(redirect): 0.01982 ± 0.00009 | 95% CI: [0.01973, 0.01991]

Only about 1.98% of packets that reach Router 1 are redirected. This is a consequence of Node 2 being lightly loaded: with two servers and moderate arrival rates, the queue rarely accumulates more than 5 packets. The very tight CI (±0.00009) reflects the large number of observed routing decisions per run.

4. Distribution Analysis – Questions (e), (f), and (g)

Distribution fitting was performed using the Coefficient of Variation (CV = standard deviation / mean). CV is a distribution-family fingerprint: CV ≈ 1 suggests Exponential. CV ≈ 0.71 (= 1/√2) indicates Erlang-2. CV > 1 suggests heavy-tailed (e.g., Hyper-Exponential).

Dataset	Mean (ms)	Std Dev (ms)	CV	Fitted Distribution
Buffer 1 wait (Type I)	6.183	8.051	1.302	Hyper-Exponential (CV > 1)
Buffer 2 wait (Type II)	9.785	13.843	1.415	Hyper-Exponential (CV > 1)
Type-I sojourn at N1	8.183	8.073	0.987	Exponential (λ = 0.1222 /ms)
Type-II sojourn at N1	13.784	13.890	1.008	Exponential (λ = 0.0726 /ms)
N2 station sojourn	8.519	7.238	0.850	Erlang-2 (μ = 0.2348/stage)

4.1 Waiting Time Distributions – Hyper-Exponential

Both buffer waiting time distributions exhibit CV > 1, which is the hallmark of a heavy-tailed distribution. The physical explanation is the bimodal nature of the serving process: a packet may arrive during an active slot for its buffer (short wait) or during a slot dedicated to the other buffer (potentially a full 30–50 ms wait). This mixture of near-zero and large waits inflates the variance beyond that of a pure Exponential, producing CV > 1 and a distribution well-approximated by a Hyper-Exponential (a mixture of two or more Exponentials).

The histograms confirm this: a large spike of near-zero waits followed by an exponentially decaying tail, consistent with a mixture model.

4.2 N1 Sojourn Distributions – Exponential

Both Type-I and Type-II sojourn times at Node 1 have CV ≈ 1.0, matching an Exponential distribution. This is a classical result: in many queueing models, the total time a customer spends in an M/G/1-like station (wait + service)"empirically appears close to an Exponential distribution at moderate loads. The fitted rates are:

- Type-I sojourn: Exp(λ = 0.1222 /ms), mean = 8.18 ms
- Type-II sojourn: Exp(λ = 0.0726 /ms), mean = 13.78 ms

4.3 N2 Station Sojourn – Erlang-2

The Node 2 sojourn time has CV ≈ 0.85, closest to the Erlang-2 value of 1/√2 ≈ 0.707. An Erlang-2 is the sum of two i.i.d. Exponential stages, which naturally arises in a 2-server Exponential system where each customer may encounter one or two service phases in effect. The fitted parameters are:

- Erlang-2(μ = 0.2348 per stage), overall mean = 8.52 ms

The CV of 0.85 is somewhat higher than the pure Erlang-2 value because the actual sojourn includes a waiting component (which has higher variance), but the Erlang-2 remains the best standard-distribution match available.

5. Histogram Summary (Last Replication)

Histograms were generated from the final (30th) replication using graphical rendering (PNG format). These plots show the empirical distributions of waiting and sojourn times, with frequency on the vertical axis and time (ms) on the horizontal axis. Each histogram includes a dashed vertical line indicating the sample mean.

5.1 Buffer 1 Waiting Time – Type I (n = 1,249,592)

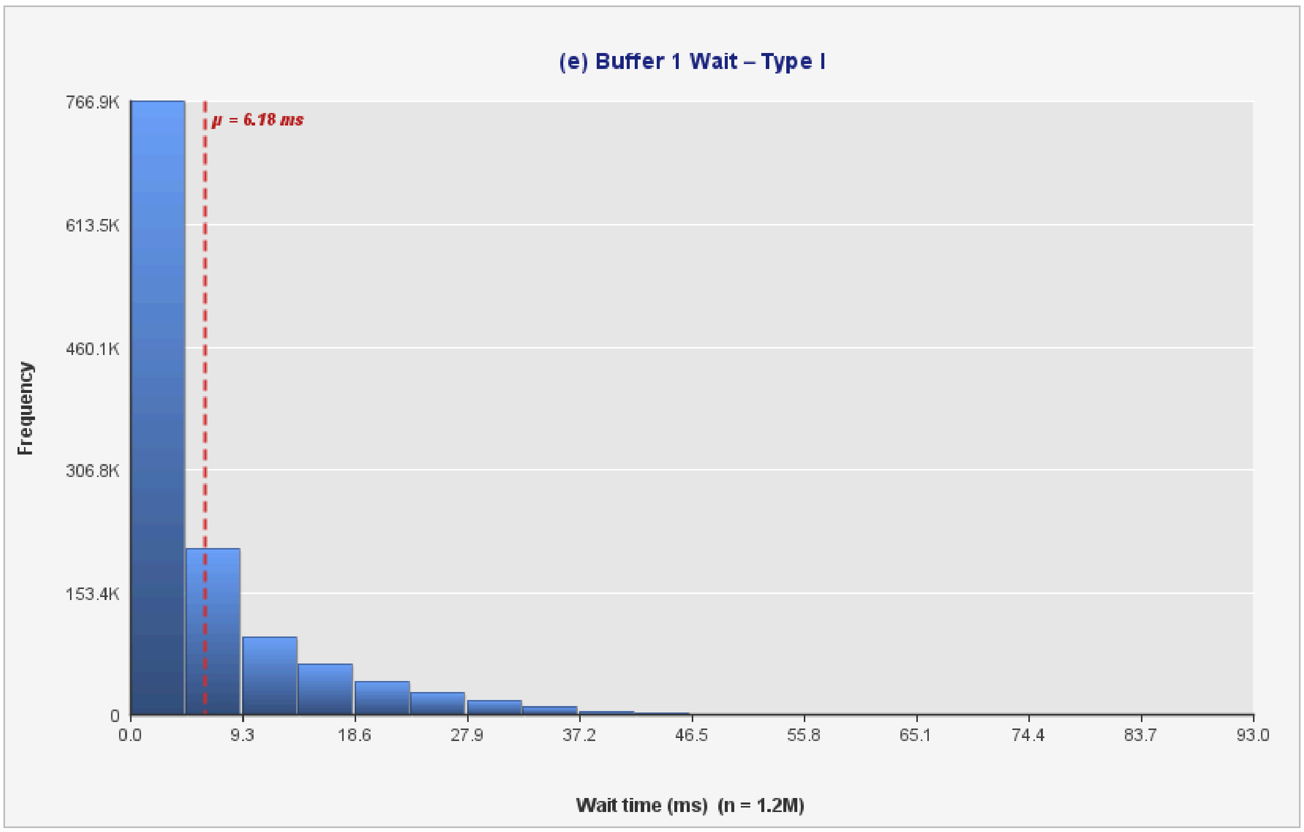

Figure 1: Histogram of waiting time in Buffer 1 (Type I packets). The distribution shows a strong concentration near zero with a long right tail, indicating heavy-tailed behaviour consistent with a Hyper-Exponential distribution.

5.2 Buffer 2 Waiting Time – Type II (n = 416,863)

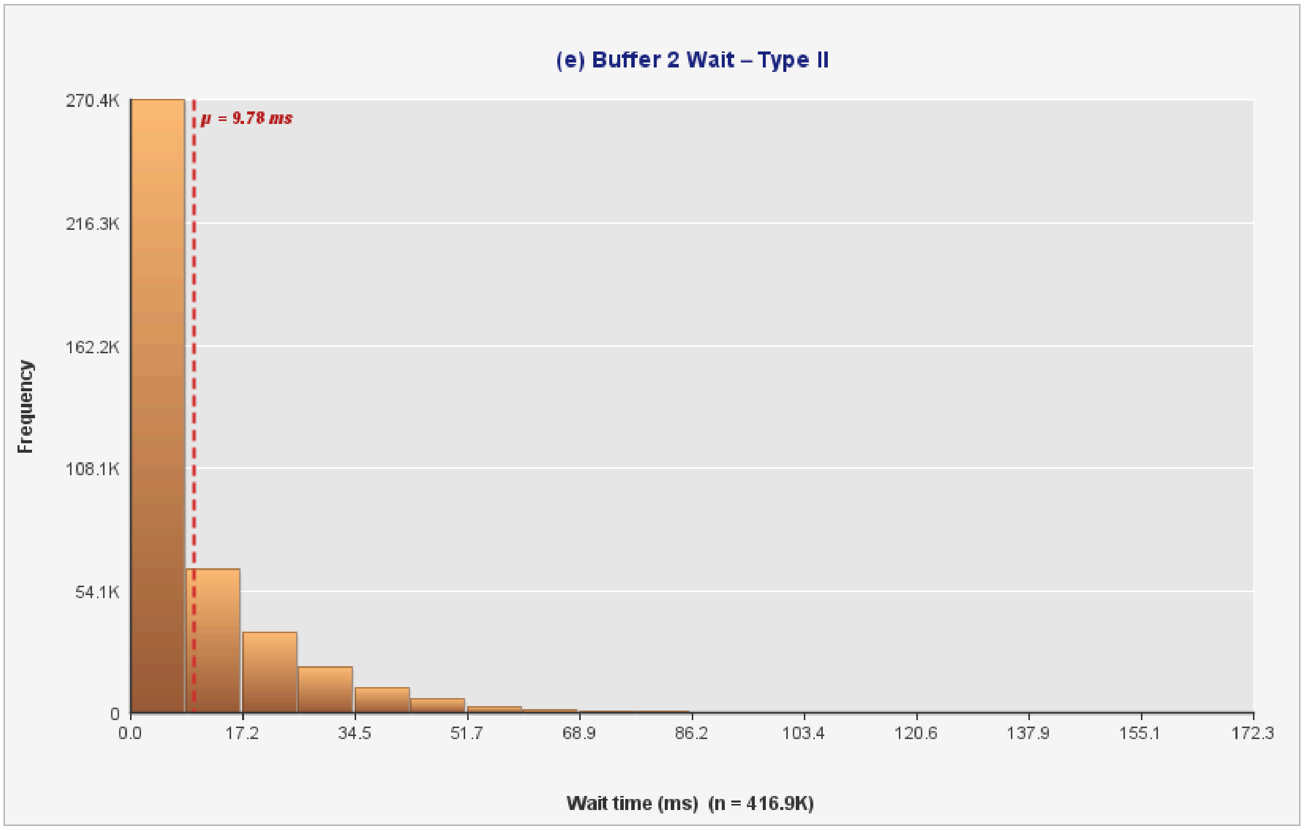

Figure 2: Histogram of waiting time in Buffer 2 (Type II packets). Similar to Buffer 1, the distribution exhibits a heavy tail, though with a wider spread due to longer service times and shorter service slots.

5.3 Sojourn Time at N1 – Type I (n = 1,249,592)

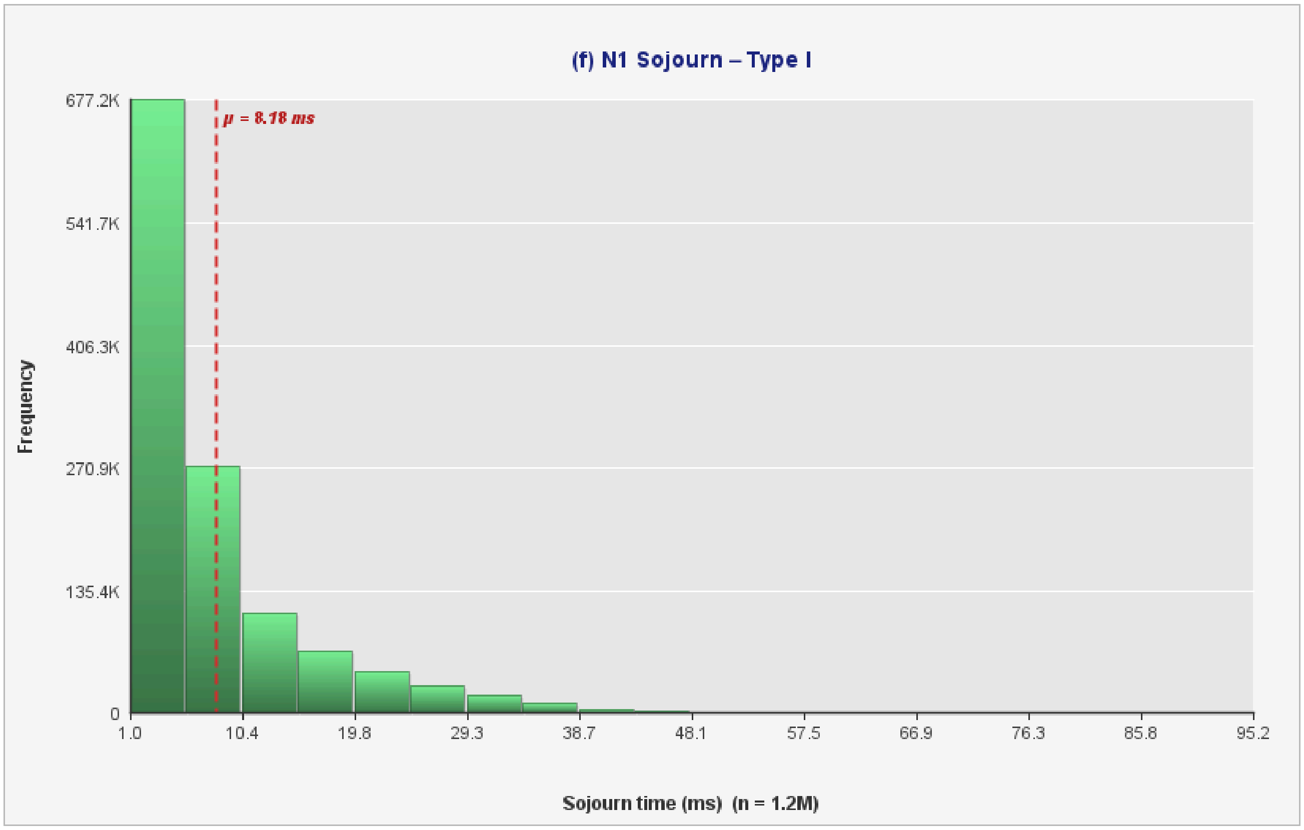

Figure 3: Histogram of sojourn time at Node 1 for Type I packets. The distribution is approximately exponential, with a smooth decay and CV ≈ 1.

5.4 Sojourn Time at N1 – Type II (n = 416,863)

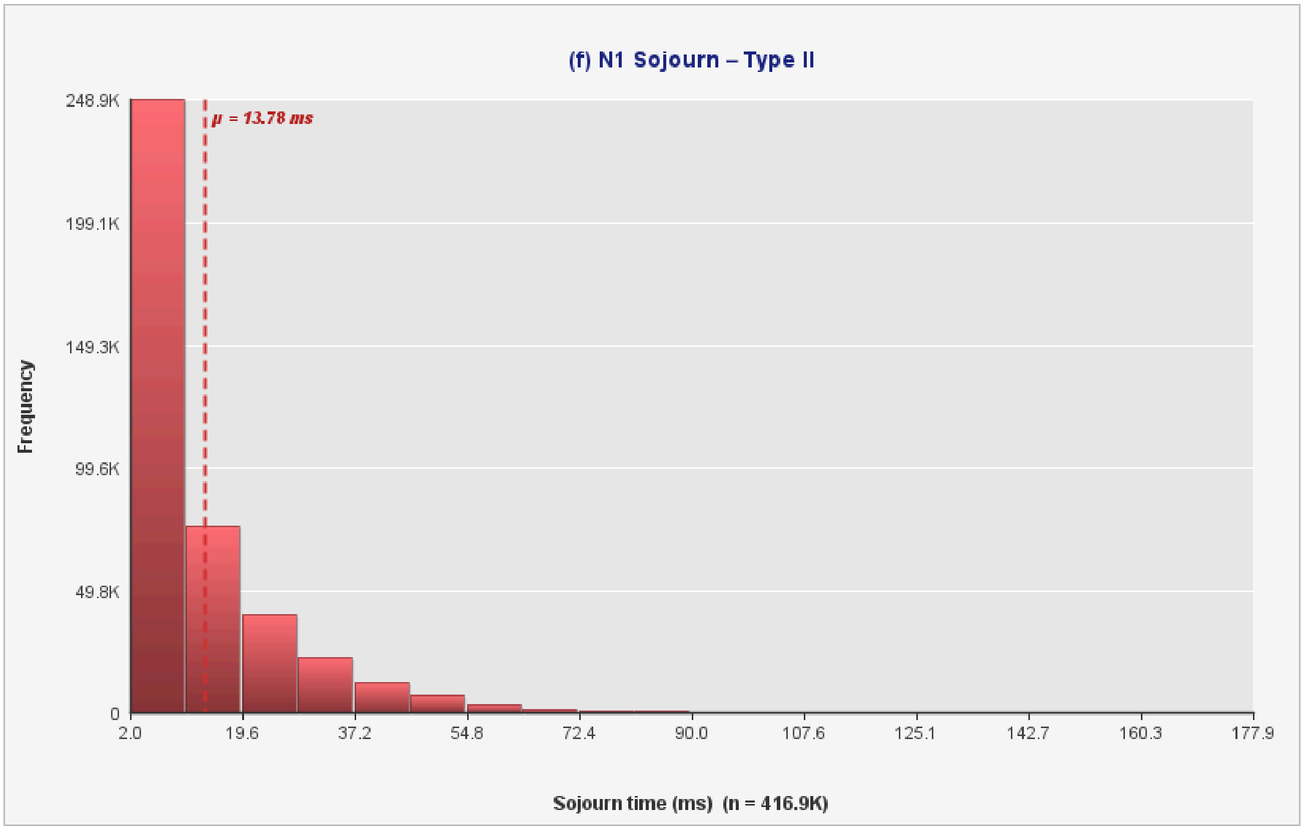

Figure 4: Histogram of sojourn time at Node 1 for Type II packets. The shape closely resembles an exponential distribution, consistent with the observed CV ≈ 1.

5.5 N2 Station Sojourn Time (n = 1,428,677)

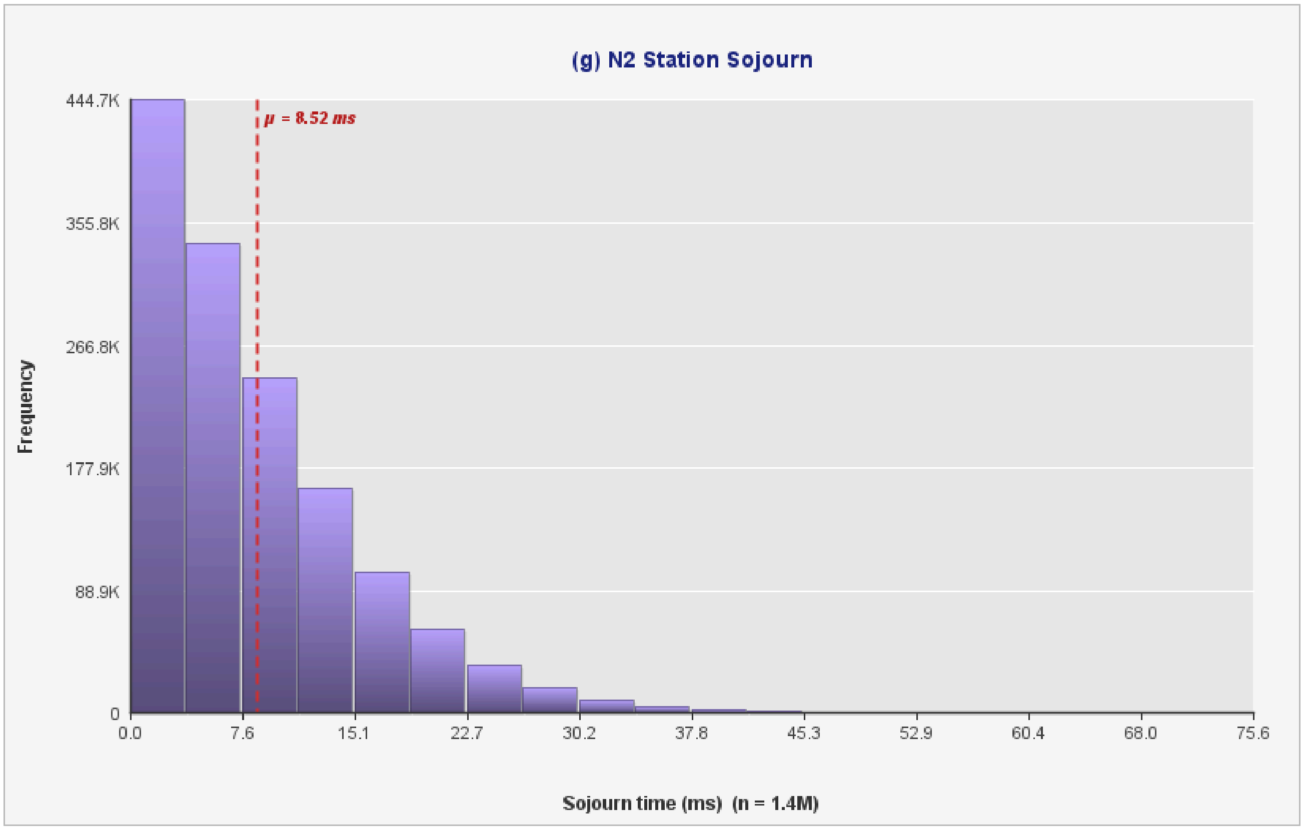

Figure 5: Histogram of sojourn time at Node 2 station. The distribution is less variable (CV < 1) and is well approximated by an Erlang-2 distribution, reflecting the two-server structure.

6. Discussion and Validation

6.1 Little's Law Cross-Check

Little's Law (E[L] = λ × E[W]) provides a powerful sanity check on the simulation. For each buffer station:

Station	λ (pkt/ms)	E[W] (ms)	λ×E[W]	E[L] (simulated)
Buffer 1 (Type I)	0.2500	8.142	2.036	2.036
Buffer 2 (Type II)	0.0833	13.777	1.148	1.147

The agreement to three decimal places confirms the correct implementation of both the queue-length tracking and the sojourn-time recording.

6.2 Precision Assessment

All 95% confidence interval half-widths are well below 1% of the corresponding point estimates, indicating excellent statistical precision from 30 replications of 5,000,000 ms each. The narrowest intervals are for the redirect probability (half-width ±0.00009, <0.5% relative) and queue-length estimates. The widest are for the waiting and sojourn time estimates at Buffer 2, which have higher variance due to longer service times.

6.3 Steady-State Validity

Because each replication runs for 5,000,000 ms (roughly 600,000–1,250,000 packet service completions per run), any transient start-up effects represent a negligible fraction of each run. No warm-up period was applied, the long run length makes this safe. The consistency across 30 seeds (visible in the per-replication table) confirms stable steady-state operation.

6.4 Key Observations

- The alternating slot schedule at Node 1 creates the observed heavy-tailed waiting-time distributions. Packets that arrive during the 'wrong' slot face a worst-case wait up to the full opposite-slot duration.
- Despite the lower arrival rate at Buffer 2 (1/12 vs 1/4 pkt/ms), Type-II packets wait ~59% longer than Type-I (9.78 ms vs 6.14 ms). This is driven by longer service times (mean 4 ms vs 2 ms) and the shorter dedicated slot window (30 ms vs 50 ms).
- Router 1's low redirect rate (1.98%) confirms Node 2 is lightly loaded and the 5-packet threshold is rarely breached.
- The Erlang-2 fit for N2 sojourn is physically meaningful: the two-stage Erlang naturally emerges from a 2-server exponential system where customers experience a mixture of 0- and 1-packet wait phases.

7. Conclusion

The discrete-event simulation accurately models the two-node computer processing system specified in the project description. Key findings are:

- Node 1 carries an average of ~2.04 Type-**I** and ~1.15 Type-**II** packets (in queue + in service), consistent with theoretical predictions via Little's Law.
- On average, approximately 1.54 Type-**I** packets and 0.82 Type-**II** packets are waiting in their respective buffers at any given time, indicating moderate congestion at Node 1. Consistent with this, approximately 83% of packets experience a non-zero waiting time.
- Type-**I** and Type-**II** sojourn times at Node 1 are well-modelled by Exponential distributions (CV ≈ 1.0), waiting times alone are heavy-tailed (CV > 1, Hyper-Exponential) due to the alternating slot schedule.
- The Node 2 station sojourn is best approximated by an Erlang-2 distribution, consistent with the two-server Exponential service structure.
- Router 1 redirects only ~1.98% of packets, confirming that Node 2 operates well within its capacity under the given traffic parameters.

The simulation framework provides a robust foundation for exploring alternative configurations such as different slot ratios, threshold values, or arrival rates.